MIL-H-2849D
23 June 1980
SUPERSEDING
MIL-H-2849C
1 March 1973

MILITARY SPECIFICATION

HOOD, ANTIFLASH

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers the requirements for one type and size knitted cotton hood.

2. APPLICABLE DOCUMENTS

* 2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-276	- Thread, Cotton
JJ-W-155	- Webbing, Textile, (Cotton, Elastic)
DDD-L-20	- Label; for Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	- Boxes, Shipping, Fiberboard

STANDARDS

* Beneficial comments (recommendations, additions, and deletions) and any pertinent data which may be of use in improving this document should be addressed to Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and STorage

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U.S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

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3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3).

3.3 Materials

3.3.1 Basic material. The basic material shall be of a knitted 100% cotton, made with singles unbleached yarn and constructed with a birds-eye or equivalent knitted stitch type to be approved by the procuring activity. The finished fabric shall conform to the requirements shown in table I when tested as specified in 4.4.1.

Table I - Physical requirements of basic material

Characteristic	Requirements	
	Minimum	Maximum
Weight, ounces per square yard	3.7	4.5
Wales per inch	21	-
Courses per inch	40	-
Bursting strength, pounds	40	-

*3.3.1.1 Nonfibrous materials. The starch and protein content, including chloroform-soluble and water-soluble materials, shall not exceed 5.0 percent when tested as specified in 4.4.1.

3.3.2 Thread. The thread for seaming and stitching the hood shall be cotton, conforming to type IAI, ticket no. 40, 3-ply, of V-T-276.

*3.3.3 Webbing, cotton elastic. The elastic cotton webbing shall be unbleached conforming to type I, class 1 or 2, of JJ-W-155.

3.3.4 Label. Each finished hood shall have an identification label conforming to type I, class 1, of DDD-L-20 and shall bear the following inscription:

HOOD, ANTIFLASH
CONTRACT NO: DSA 100-00-000-0000 (EXAMPLE)
STOCK NO. 8415-00-000-0000 (EXAMPLE)
NAME OF CONTRACTOR:

The printing shall show fastness to laundering.

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3.4 Design. The hood shall be shoulder length, round at the top and flared at the bottom. The circular face opening shall be hemmed with an elastic webbing enclosed.

3.4.1 Figure. Figure 1 is furnished for information purposes only. When inconsistencies exist between the written specification and the figure, the written specification shall govern.

3.5 Patterns. Standard patterns to be used to cut the working patterns will be furnished by the Government (see 6.3). The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way.

3.5.1 List of pattern parts. The component parts of the hood shall be cut from the material as specified, and in accordance with the number of parts required for the manufacturing process as follows:

Materials	Nomenclature of pattern parts	Cut parts
Cloth, knitted, Cotton	Front	1
	Back	1

3.6 Construction

3.6.1 Stitches, seams, and stitchings. Stitch, seam, and stitching types as specified in Table II, shall conform to the requirements of FED-STD-751. Whenever two or more methods for seams or stitchings are given for the same operation, any one of them may be used.

3.6.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table II.

3.6.3 Thread breaks and ends of seams. Ends of all seams and stitchings, when not caught in other seams or stitchings, shall be backtacked not less than 1/2 inch. Thread breaks (all stitch types) shall be secured by stitching back of break not less than 1/2 inch. Skipped stitches or thread breaks, may be repaired by using 301 stitch type.

3.7 Manufacturing operation requirements. The contractor shall manufacture the hoods in accordance with the operations specified in Table II. The contractor is not required to follow the exact sequence of operations listed.

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Table II - Construction of hood

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
1.	Cutting a. Cut the hoods in strict accordance with patterns furnished, which shall show size, shape, and directional lines for proper assembly of respective parts. b. Cut the webbing for the face opening, 19 (± 1/4) inches in length (see 3.3.3).					
2.	Replacement of defective components During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in Section 4, shall be removed from production and replaced by non-defective and properly matched components.					
3.	Hem face opening a. Overlap ends of the cotton webbing 7/8 (± 1/8) inch, and tack or bartack across its width at center of the lap.	301 or bartack	LSa-1	10-12 28 per bartack	40/3	40/3
	b. Stitch an edge of the cotton webbing to the inside raw edge of face opening	503 or 504	SSa-1	10-12	40/3	40/3
	c. Turn under the webbing 1/2 inch, i.e., the width of the webbing, and stitch to the body of hood 1/16 to 1/8 inch from the sewn edge.	301	EFa-1	10-12	40/3	40/3

NCTR Form 1, 15 Sep 1967

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Table II - Construction of hood (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Join front and back Join front to back face to face, stitching along the sides and around the top in a continuous operation.	515 or 516 or 517 or 518 or 519	SSa-2	10-12	40/3	40/3
5.	Hem lower edge of hood and attach label a. Fold under lower or bottom edge of hood 1/2 (± 1/8) inch to form an inside hem, and stitch continuously around its circumference simultaneously positioning and catching the bottom edge of label at center back in the line of stitching.	503 or 504	EFc-1	10-12	40/3	40/3
	b. Stitch remaining three sides of label to hood, 1/8 inch from the edge. or * As an alternate, label may be positioned at center back 1/8 inch above hem and stitched on four sides of label to hood, 1/8 inch from edge.	301 301	ISbj-1 LSbj-1	10-12 10-12	40/3 40/3	40/3 40/3
	c. The label shall finish face side up, facing towards the top of the hood with all of the printing clearly visible.					
6.	Clean hoods a. Trim all thread ends b. Remove all loose threads c. Remove all spots and stains.					

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3.8 Measurements. The finished hood shall conform to the dimensions specified in table III. All measurements are expressed in inches, positioned as follows:

- Place the hood smooth and flat, face side up with the bottom edges even

Table III - Finished measurements

Measurement		Tolerance
Overall length (A)	19	± 1/2
Diameter of face opening (B)	5 1/4	± 1/2
Width at base of face opening (C)	12 3/4	± 1/2
Width at widest point of hood (D)	22 1/2	± 1/2

NOTE: A through D refer to figure 1.

3.9 Workmanship. The finished hoods shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels..

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection When required, the first article submitted in accordance with 3.2 shall be inspected for compliance with design, construction, workmanship, appearance, and dimensional requirements.

4.4 Quality conformance inspection Inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

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4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition, the knitted basic material shall be tested for the characteristics specified in table IV and in accordance with the referenced test methods of FED-STD-191 whenever applicable. Testing for all the requirements listed is applicable to the sample unit. The lot size shall be expressed in units of yards and the sample unit shall be 1 yard full width. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table IV - Fabric tests

Component	Characteristic	Requirement reference	Test method
Cloth, knitted, cotton	Material identification	3.3.1	1/
	Type of knit	3.3.1	1/
	Weight	Table I	5041
	Wales and courses per inch	Table I	5070
	Bursting strength	Table I	5120
	Nonfibrous material	3.3.1.1	2611

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4.4.2 Examination of the end item Examination of the end item shall be in accordance with 4.4.2.1 and 4.4.2.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.4.2.3. The sample unit shall be one finished hood. The lot size shall be expressed in units of hoods.

4.4.2.1 Visual examination The hoods shall be examined for defects classified in accordance with the list below:

Defect	Classification Major	Minor
MATERIAL DEFECTS AND WORKMANSHIP DAMAGES:		
a. Any hole, needle chew, cut, tear or burn	X	
b. Weakening defect such as slub, smash or float	X	
c. Any visible mend	X	
d. Any slubby or heavy yarn		X
e. Sleazy or boardy hand		X

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Defect	Classification Major	Minor
CLEANNESS:		
a. Any spot or stain		X
b. Thread ends not trimmed or loose thread not removed		X
COMPONENTS AND ASSEMBLY:		
Any component part or operation omitted or not as specified (unless otherwise classified herein)	X	
CUTTING:		
Any part not cut in accordance with directional lines indicated on patterns or in accordance with specification requirements	X	
COLOR:		
Other than unbleached		X
SEAMS AND STITCHING:		
a. Accuracy of seaming:		
1. Seams puckered, twisted or pleated (unless otherwise classified herein)		X
2. Ends of all seams and stitching when not caught in other seams or stitchings back-tacked less than 1/2 inch		X
3. Thread breaks not secured or stitched back of the break less than 1/2 inch		X
4. Part of hood caught in any unrelated stitching operation, seriously affecting serviceability	X	
b. Gauge of stitching:		
1. Irregular, i.e., unevenly gauged, not uniform		X
2. Beyond range of width specified		X
c. Open seams or runoffs:		
1. Up to 1/4 inch inclusive		X
2. More than 1/4 inch	X	

NOTE: One or more broken stitches or two or more consecutive skipped or runoff stitches constitute an open seam.

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Defect	Classification Major	Minor
SEAMS AND STITCHING: (cont'd)		
d. Raw edges:		
1. More than 1/4 inch to and including one inch		X
2. More than one inch	X	
e. Seam and stitch types:		
1. Not specified seam or stitch type		X
f. Stitch tension:		
1. Loose tension resulting in loose seam	X	
2. Tight tension (stitches break when normal strain is applied to the seam or stitching)	X	
g. Stitches per inch:		
1. One stitch less than the minimum or any number in excess of the maximum specified		X
2. Two or more stitches less than the minimum specified	X	
LABEL		
a. Missing, incorrect, or illegible	X	
b. Misplaced		X
c. Not type specified		X

5. PACKAGING

5.1 Preservation-packaging Preservation-packaging shall be level A of commercial as specified (see 6.2).

5.1.1 Level A

5.1.1 Unit packaging Each hood shall be laid flat, face side up. The right and left bib portions shall be folded inward so that the sides are comparatively straight. The hood shall then be folded in half from top to bottom, to finish approximately 13 inches x 7 1/2 inches. Twenty (20) hoods shall be neatly stacked in a bundle, with each bundle securely tied with cotton tape or twine approximately 2 inches from each end.

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5.1.2 Commercial Hoods shall be preserved-packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A Two hundred and forty (240) hoods, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to type SF, class weather-resistant grade V2s, style CSOSC, of PPP-B-636. Level A packages shall be packed flat, two in length, two in width, and three in depth within the shipping container. Inside dimensions of each container shall approximate 26 1/2 inches in length, 15 1/2 inches in width and 15 inches in depth. Each container shall have the contents completely covered on the top and bottom with a sheet of kraft paper of any suitable commercial grade, closed in accordance with Method III, and reinforced in accordance with the appendix of PPP-B-636.

5.2.2 Level B Two hundred and forty (240) hoods, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style CSOSC, type CF (variety SW) or SF, class domestic, grade 200, of PPP-B-636. Level A packages shall be packed flat, two in length, two in width, and three in depth within the shipping container. Inside dimensions of each container shall approximate 26 1/2 inches in length, 15 1/2 inches in width, and 15 inches in depth. Each container shall have the contents completely covered on the top and bottom with a sheet of kraft paper of any suitable commercial grade, and shall be closed in accordance with Method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard containers When specified (see 6.2), the shipping containers shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with Method III as specified in the appendix of the box specification.

5.2.3 Commercial Hoods, preserved-packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall be in accordance with US Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

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6.1 Intended use. The hoods covered by this specification are intended to protect military personnel from transient and elevated air temperatures, resulting from the use of high explosive weapons.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When first article sample is required (see 3.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
d. When weather resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids (see 3.1).

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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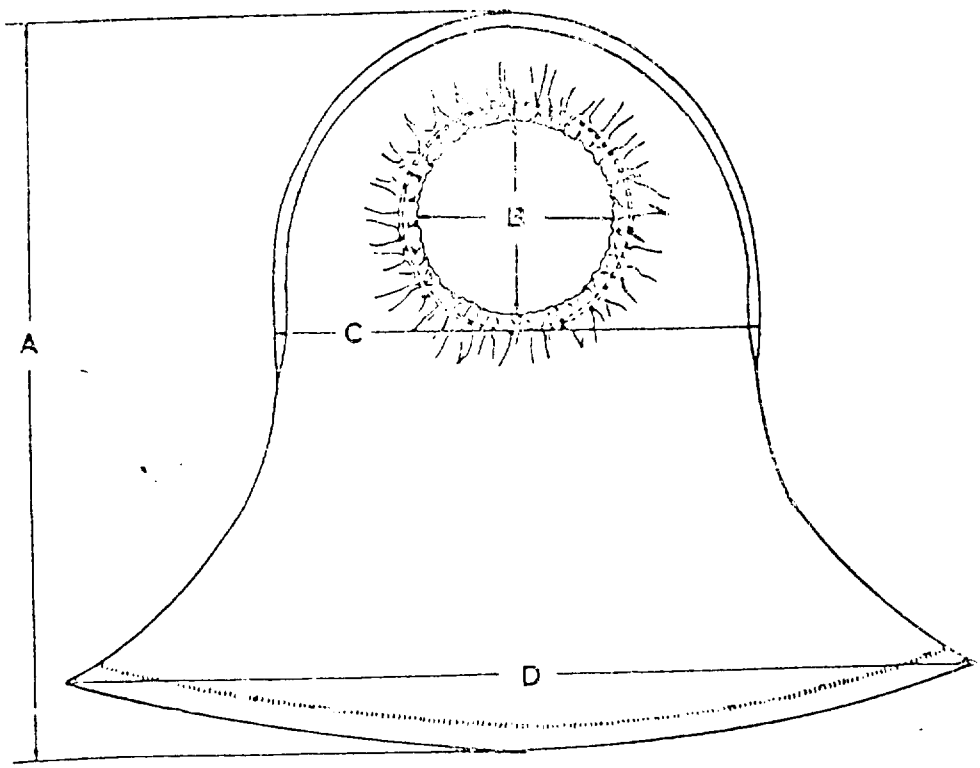
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GURE 1 - HOOD, ANTIFLASH

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